

Specialist Machine Learning for Cheese Shelf-Life Prediction

Modeling Strategy, Validation, Explainability, and Application Integration

Shelf-Life Studio — Cheese Shelf-Life Prediction, Classification, and Ingredient Ranking System

Technical research article based on the corrected V7 specialist pipeline and the current application architecture

Abstract — Shelf-Life Studio is a machine-learning decision-support platform for cheese shelf-life analysis. The central design decision is to avoid a single global regression model and instead route each case to one of six specialists defined by cheese category (soft, semi-hard, hard) and prediction task (general shelf life or safety endpoint). Each specialist independently trains Random Forest, LightGBM, XGBoost, and Explainable Boosting Machine (EBM) regressors under a leakage-safe context split, with production selection based exclusively on validation RMSE. On the corrected V7 data, general shelf-life specialists achieve strong held-out performance: test $R^2 = 0.971$ for soft, 0.920 for semi-hard, and 0.932 for hard cheese. Safety-endpoint performance is substantially weaker (0.594, 0.317, and -0.011 , respectively), but this degradation tracks the proportion of right-censored lower-bound targets: 69%, 93%, and 100%. The platform therefore treats safety predictions as a separate statistical problem and explicitly reports support limitations. The serving layer additionally performs out-of-distribution checks, matched control-versus-treatment comparison, residual-based uncertainty intervals, exact SHAP/EBM local decomposition, training-range visualization, and what-if sensitivity analysis. Complementary systems classify full formulation efficacy and rank ingredients using context-adjusted effects. This article documents the end-to-end modeling strategy, numerical results, limitations, and implementation logic needed to defend the system scientifically.

Keywords — cheese shelf life; gradient boosting; Random Forest; Explainable Boosting Machine; SHAP; right censoring; challenge testing; conformal uncertainty; food preservation; interpretable machine learning.

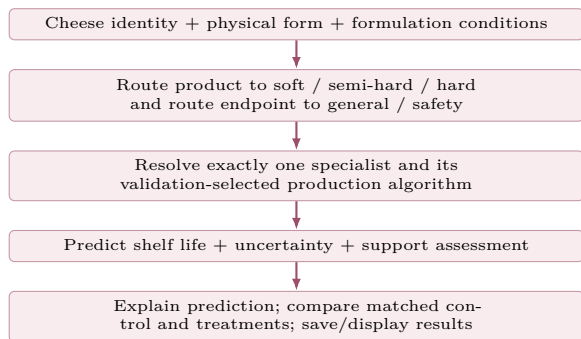


Figure 1. Production logic of the specialist regression system. No safety-to-general or category-to-category fallback is permitted when a specialist is unavailable.

1. Problem formulation and system objective

Shelf life is not an intrinsic constant attached to an ingredient or cheese name. It is the result of an experimental context: matrix composition, physical presentation, storage temperature, packaging atmosphere, processing history, endpoint definition, treatment, dose, and application method. The primary regression objective is to predict the number of days until the selected endpoint is reached. For an input formulation/context vector x , the production model is

$$\hat{y} = f_{c,t}(x), \quad (1)$$

where $c \in \{\text{soft, semi-hard, hard}\}$ and $t \in \{\text{general, safety}\}$. The notation is deliberate: the application does not assume one universal function for every cheese and endpoint.

The application contains three complementary predictive systems. The first is the six-specialist shelf-life regression architecture described in this paper. The second classifies a complete treated formulation into Low/Medium/High efficacy relative to a matched control. The third ranks individual ingredients by their typical context-adjusted shelf-life effect. An AI assistant sits above these services as an orchestration layer; it is not allowed to invent project-specific values and must obtain predictions, metrics, explanations, and rankings through the same backend services used by the application.

1.1 End-to-end architecture

2. V7 data representation and feature design

The corrected V7 training corpus contains 34,000 modeled rows across three category-specific files. Each category is partitioned into general shelf-life and safety-endpoint subsets before model training. The six resulting specialist to-

als are 8,800 (soft/general), 4,800 (soft/safety), 7,200 (semi-hard/general), 4,000 (semi-hard/safety), 6,000 (hard/general), and 3,200 (hard/safety).

The feature space is intentionally scientific rather than purely statistical. Matrix variables include food matrix, pH, water activity, moisture, fat, protein, salt, ripening time, and physical form. Processing/storage variables include pasteurization, storage temperature, packaging type, and headspace gases. Endpoint variables include indicator type/group, threshold, threshold unit, and initial indicator value. Treatment variables include treatment type, application method, ingredient identity/family, ingredient count, and concentration.

Concentration required special handling because values can be reported in fundamentally different units. The live form accepts familiar raw units, while the backend converts them into a canonical value/unit representation before constructing the model row. This prevents a numeric column from incorrectly treating ppm, mg/kg, percent w/w, percent w/v, IU/g, and log CFU/g as if they shared one physical scale.

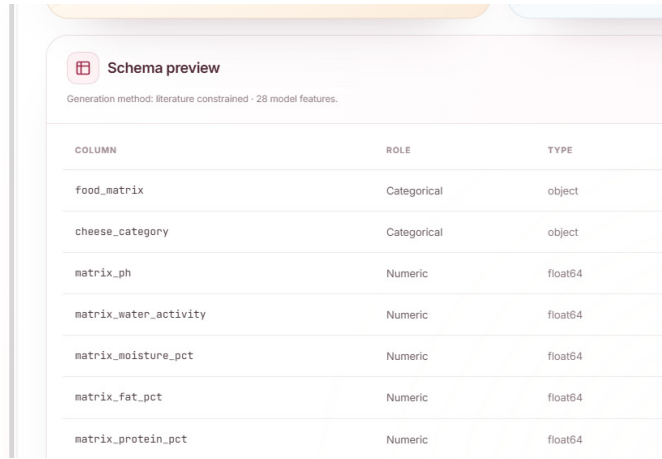
Safety specialists also include challenge-context variables such as initial inoculum (log CFU/g) and microbial growth support. Rather than hard-code these fields as “safety-only”, the training script dynamically excludes any feature that is 100% null within a specialist’s task-filtered data. The resulting schema contains 28 modeled features for each general specialist and 30 for each safety specialist.

Physical form is introduced during product identification because the same base cheese can appear in distinct commercial forms and therefore route differently in the catalog. In the current implementation it is also retained as contextual model information and as part of support assessment; a live request is never allowed to silently guess a missing physical form.

Identifiers and provenance fields are excluded from training. In particular, row/context/formulation identifiers, source URLs, rule/version metadata, audit columns, and target-recalibration fields are not allowed to become predictors. The target itself, `shelf_life_days`, is asserted not to appear in the feature schema. This separation is critical because provenance may correlate with the target while being unavailable or scientifically meaningless at prediction time.

3. Leakage-safe split and training protocol

A row-level random split would be unsafe because several rows can belong to the same experimental context. Treatment and control rows sharing a context may differ only in preservation variables. If one were placed in training and the other in validation/test, the model could effectively recognize the experiment



COLUMN	ROLE	TYPE
food_matrix	Categorical	object
cheese_category	Categorical	object
matrix_ph	Numeric	float64
matrix_water_activity	Numeric	float64
matrix_moisture_pct	Numeric	float64
matrix_fat_pct	Numeric	float64
matrix_protein_pct	Numeric	float64

Figure 2. Real application capture of the Data workspace and schema preview. The paper’s numerical results are taken from the corrected V7 artifacts; interface screenshots are representative application captures from the iterative development process.

rather than generalize to a new context.

The system therefore groups by `context_id` and assigns complete contexts to train, validation, or test using seed 42. The target proportions are approximately 70%/15%/15%. The exact specialist split counts are shown in Table 1. All preprocessing objects are fitted only on the training split. Validation is used for early stopping, model selection, and uncertainty calibration. The test split remains held out for final reporting.

3.1 Preprocessing

Numeric/binary variables use median imputation where necessary. Categorical variables use a constant missing marker followed by one-hot encoding for tree models. EBM uses its own frame-oriented imputation representation. One-hot-expanded contributions are mapped back to original scientific source features for interpretation, so the user sees “Packaging” or “Storage temperature”, not implementation labels such as `cat_packaging_type_vacuum`.

4. Candidate regression algorithms

Four independently fitted regressors compete inside every specialist.

4.1 Random Forest

Random Forest averages a large collection of decorrelated regression trees [1]. The specialist configuration uses 400 trees, unrestricted depth, minimum leaf size 2, a fixed seed, and parallel fitting. It provides a robust nonlinear baseline capable of learning interactions without explicit equations.

4.2 LightGBM

LightGBM is a gradient-boosted tree system optimized for tabular data [2]. The project permits up to 2,000 boosting rounds with learning rate 0.03, 31 leaves, row/feature subsampling, and validation-based early stopping after 50 rounds without improvement. The large maximum is therefore only an upper bound; validation loss controls the effective ensemble size.

4.3 XGBoost

XGBoost is a regularized gradient-boosted tree implementation [3]. The configuration uses up to 2,000 trees, learning rate 0.03, maximum depth 6, 90% row and feature subsampling, and early stopping. It is retained alongside LightGBM because the two optimization/tree-growth strategies can behave differently on the same specialist.

4.4 Explainable Boosting Machine

EBM is a generalized additive model with learned nonlinear univariate functions and a limited number of pairwise interactions [4]:

$$\hat{y} = \beta_0 + \sum_j f_j(x_j) + \sum_{(j,k) \in \mathcal{I}} f_{jk}(x_j, x_k). \quad (2)$$

The implementation allows 10 interactions, 256 bins, learning rate 0.02, up to 3,000 rounds, minimum leaf/bin support 4, and outer bagging. EBM is particularly useful because its prediction is intrinsically decomposable into an intercept and explicit term contributions.

5. Metrics, selection rule, and uncertainty

The training pipeline records train, validation, and test R^2 , RMSE, MAE, median absolute error, and MAPE. The principal metrics are

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|, \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}, \quad (4)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (5)$$

Model selection is based only on the lowest *validation* RMSE. Test performance never determines which algorithm is deployed. This is essential: selecting on the test set would convert the test set into another tuning set.

For uncertainty, the absolute validation residuals are computed for each saved model and their 90th percentile $q_{0.90}$ is stored. A live prediction is returned with

$$[\max(0, \hat{y} - q_{0.90}), \max(0, \hat{y} + q_{0.90})]. \quad (6)$$

This is a split-conformal-style empirical residual interval [6]. It should be interpreted as a validation-calibrated uncertainty band under approximately similar data conditions, not as a guarantee under strong distribution shift.

6. Headline V7 specialist results

Table 1 reports the production algorithm and held-out performance for each specialist. General shelf-life models are consistently strong, whereas safety specialists degrade as target censoring becomes dominant.

6.1 General shelf-life models

The general task is the strongest component of the regression system. Soft/general reaches test $R^2 = 0.971$ with MAE 2.53 d. Its validation and test R^2 values are nearly identical (0.972 and 0.971), and the train-validation gap is only 0.025. Semi-hard/general reaches test $R^2 = 0.920$ with MAE 8.00 d and gap 0.059. Hard/general reaches test $R^2 = 0.932$ with MAE 6.93 d and gap 0.076. In all three categories, LightGBM is selected by validation RMSE.

The correct interpretation is not simply that one error in days is “better” than another across categories. Soft products often occupy shorter target scales than hard cheeses; therefore MAE/RMSE must be interpreted alongside the target range and R^2 . The consistency between validation and test is more informative than comparing absolute errors across distinct shelf-life distributions.

6.2 Safety endpoints and the censoring problem

Safety endpoints are statistically different from ordinary spoilage/quality endpoints. A general shelf-life study commonly observes a failure event—for example sensory rejection or a spoilage threshold—and records an actual time. In a pathogen challenge study, the critical microbial threshold may never be reached before the experiment ends. If a study stops at day T with no threshold crossing, the information is

$$T_{\text{safety}} > T, \quad (7)$$

not $T_{\text{safety}} = T$. The target is a right-censored lower bound.

V7 explicitly records this situation through `target_is_lower_bound`. The test censoring fractions are 69% for soft/safety, 93% for semi-hard/safety, and 100% for hard/safety. Test R^2 declines in the same order: 0.594, 0.317, and -0.011. This is not evidence that censoring explains

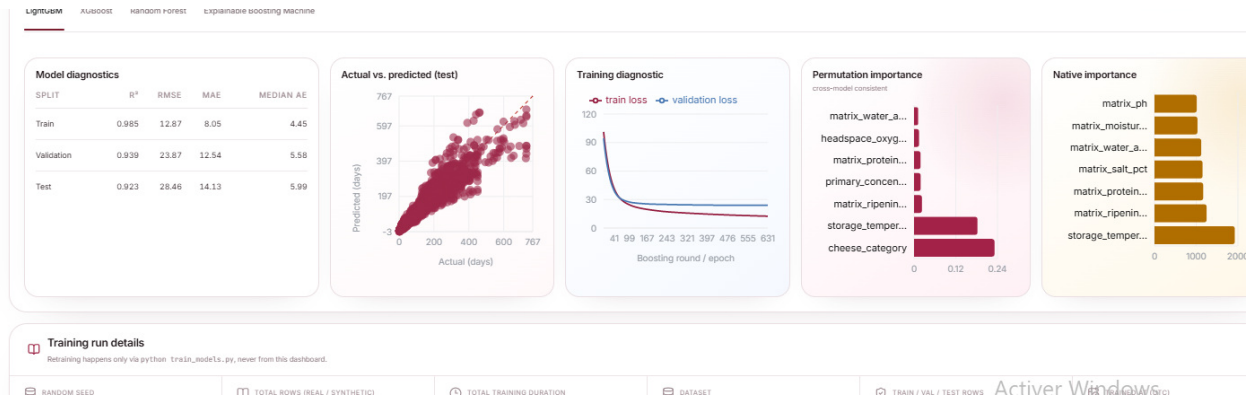


Figure 3. Real application capture of the modeling diagnostics workspace: split-wise metrics, actual-versus-predicted behavior, training diagnostics, and feature importance. The current backend now scopes these diagnostics to the selected category/task specialist.

Table 1. Current corrected V7 production specialist results. Production model = lowest validation RMSE. LB = right-censored lower-bound target.

Specialist	Best	n_{tr}	n_{val}	n_{te}	Val R^2	Test R^2	Test MAE	Test RMSE	Gap	LB test
Soft / General	LightGBM	6160	1320	1320	.972	.971	2.53	4.89	.025	0%
Soft / Safety	XGBoost	3360	720	720	.592	.594	7.73	10.34	.239	69%
Semi-hard / General	LightGBM	5040	1080	1080	.931	.920	8.00	13.39	.059	0%
Semi-hard / Safety	XGBoost	2800	600	600	.115	.317	13.26	16.05	.353	93%
Hard / General	LightGBM	4200	900	900	.916	.932	6.93	11.30	.076	0%
Hard / Safety	XGBoost	2240	480	480	.008	-.011	17.80	20.98	.038	100%

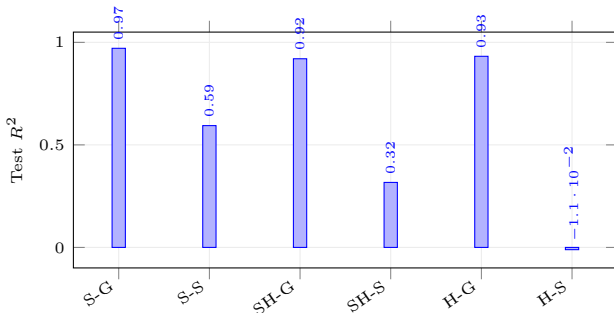


Figure 4. Held-out R^2 for each production specialist. S=soft, SH=semi-hard, H=hard; G=general shelf life, S=safety endpoint.

every error, but it establishes a clear identifiability limitation: ordinary squared-error regression treats “at least T ” as if it were an exact target.

This also explains why censoring appears primarily in safety rather than general shelf life. General shelf-life studies are usually designed to observe spoilage/quality failure within the study window. Safety challenge tests can terminate while the pathogen is still below the critical criterion; the absence of failure is itself a scientifically useful outcome but does not reveal the exact event time.

6.3 Hard/safety collapse

The hard/safety production XGBoost model has train $R^2 = 0.046$, validation $R^2 = 0.008$, and test $R^2 = -0.011$. Inspection of saved predictions shows true training-target standard deviation 21.06 d but predicted standard deviation only 0.77 d—3.6% of the target variation. The model has effectively collapsed toward a near-constant prediction. This is consistent with a task in which all held-out targets are lower bounds. The correct scientific response is not to hide this result or multiply outputs by a correction factor; it is to treat safety as a censoring-aware time-to-event problem in future work.

7. Full algorithm comparison

Tables 2 and 3 show that the production choices are not arbitrary. For general shelf life, LightGBM has the best validation RMSE in each category, although XGBoost is extremely close and even obtains a marginally better hard-cheese test R^2 ; it is not selected because test performance is not a tuning criterion. For safety, XGBoost has the lowest validation RMSE in all three categories.

8. Live prediction and matched-treatment comparison

The prediction UI constructs a row in stages: product identity and physical form; physicochemical profile; endpoint; storage/packaging; treatment; and concentration. Endpoint routing is read from the trained specialist schemas rather than duplicated as an independent frontend rule. Once category and task are resolved, exactly one specialist service is used and its previously selected best algorithm performs the prediction.

For treatment comparison, the application builds a matched control row that keeps the shared matrix, physical form, storage, packaging, and endpoint conditions unchanged while replacing treatment-related fields with the control template. Candidate j is summarized by

$$\Delta_j = \hat{y}_j - \hat{y}_0, \quad (8)$$

$$I_j = 100 \frac{\hat{y}_j - \hat{y}_0}{\hat{y}_0}, \quad (9)$$

$$R_j = \frac{\hat{y}_j}{\hat{y}_0}, \quad (10)$$

where $\hat{y}_0$ is the matched control prediction. If $\hat{y}_0$ is essentially zero, the relative improvement and ratio are deliberately left undefined rather than producing an unstable huge percentage. Candidate ranking uses predicted candidate shelf life as the primary ordering variable and absolute improvement as a secondary key.

Negative raw model predictions are clipped to zero because negative shelf life is physically meaningless. This is a physical-output constraint, not a calibration mechanism: no multiplier, offset, or manual correction is used to improve reported performance.

9. Extreme cases and support assessment

The system distinguishes “a model can numerically return a value” from “the value is supported by its training domain.” Each live request receives a structured support assessment.

Categorical values are checked against the exact specialist training vocabulary. An unseen category is reported explicitly rather than silently encoded as if it were ordinary. Numeric inputs are compared with training ranges. Where a pooled numeric range would mix incompatible contexts, the system prefers a range conditioned on relevant categorical variables. This matters for variables such as indicator thresholds and concentration, whose meaningful ranges depend on units and endpoint context.

Table 2. All algorithms — general shelf-life specialists. Values are days for MAE/RMSE. Bold rows are validation-selected production models.

Category	Algorithm	Train R^2	Val R^2	Test R^2	Val MAE	Test MAE	Val RMSE	Test RMSE
Soft	Random Forest	.997	.955	.956	2.84	2.95	5.67	5.99
Soft	LightGBM	.997	.972	.971	2.39	2.53	4.49	4.89
Soft	XGBoost	.999	.969	.968	2.38	2.55	4.65	5.11
Soft	EBM	.985	.944	.944	3.89	3.88	6.32	6.79
Semi-hard	Random Forest	.995	.925	.905	8.34	9.22	13.42	14.57
Semi-hard	LightGBM	.990	.931	.920	7.35	8.00	12.82	13.39
Semi-hard	XGBoost	.995	.930	.927	7.50	7.81	12.99	12.74
Semi-hard	EBM	.983	.900	.909	9.65	9.13	15.49	14.30
Hard	Random Forest	.994	.910	.927	8.48	7.82	12.68	11.68
Hard	LightGBM	.991	.916	.932	8.03	6.93	12.26	11.30
Hard	XGBoost	.995	.911	.933	8.05	6.96	12.55	11.26
Hard	EBM	.977	.880	.917	9.43	8.31	14.58	12.47

Table 3. All algorithms — safety-endpoint specialists. The weak semi-hard/hard results should be interpreted with the censoring fractions in Table 1.

Category	Algorithm	Train R^2	Val R^2	Test R^2	Val MAE	Test MAE	Val RMSE	Test RMSE
Soft	Random Forest	.997	.548	.572	8.62	7.85	10.97	10.62
Soft	LightGBM	.876	.583	.573	8.43	7.85	10.54	10.61
Soft	XGBoost	.832	.592	.594	8.31	7.73	10.42	10.34
Soft	EBM	.944	.393	.360	10.10	10.48	12.72	12.98
Semi-hard	Random Forest	.997	.015	.342	15.43	12.93	18.26	15.76
Semi-hard	LightGBM	.449	.089	.303	15.06	13.60	17.56	16.21
Semi-hard	XGBoost	.468	.115	.317	14.81	13.26	17.30	16.05
Semi-hard	EBM	.936	-.280	-.127	17.00	17.17	20.81	20.63
Hard	Random Forest	.996	-.131	-.097	17.96	18.28	21.22	21.86
Hard	LightGBM	.036	-.003	-.024	17.14	17.90	19.98	21.12
Hard	XGBoost	.046	.008	-.011	17.00	17.80	19.87	20.98
Hard	EBM	.953	-.731	-.528	20.94	21.00	26.25	25.79

The registry combines model support with specialist availability, physical-form support, and sample-size warnings. Headline levels include supported, extrapolation, unsupported categorical, weak physical-form support, thin sample, and unavailable. A missing specialist is a hard failure: safety cannot silently fall back to general shelf life, and one cheese category cannot borrow another category’s model.

This architecture is important for extreme cases. An unusually high temperature or concentration can still be scored when technically possible, but the UI can label the result as extrapolation. A category never seen in training is more serious and is surfaced as unsupported. This behavior is preferable to producing a normal-looking number without qualification.

10. Explainability: global and local

Global feature importance answers “which variables matter across many rows?” The saved artifacts contain both native model importance where available and validation-set permutation importance. Permutation importance measures the drop in predictive performance when one feature is disrupted, while native importance is model-specific; the two should not be interpreted as identical quantities.

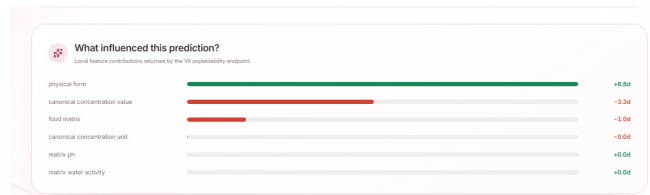
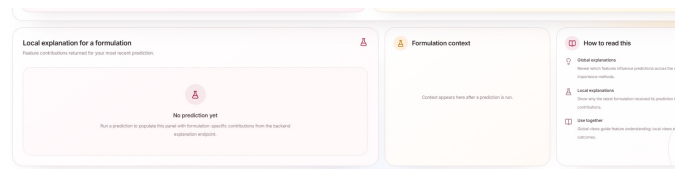
Local explainability answers the more useful question for a prediction: “why did this particular formulation receive this value?” The current detailed explanation endpoint uses the exact routed specialist and the exact algorithm that produced the prediction.

For EBM, the decomposition is native: the intercept plus local main-effect and interaction scores reconstruct the prediction. For tree ensembles (LightGBM, XGBoost, Random Forest), SHAP TreeExplainer [5] provides an additive decomposition

$$\hat{y} = \phi_0 + \sum_j \phi_j, \quad (11)$$

with one-hot contributions aggregated back to source variables. The backend verifies reconciliation on every request with a tolerance of 0.05 d. Small/zero terms are hidden; the remaining residual can be shown as “all other factors” so that the visual explanation remains mathematically honest.

For important numeric features, the app also reports where the actual input lies relative to the selected specialist’s training minimum, median, and maximum. A separate sensitivity routine varies one controllable feature—currently storage temperature, pH, or concentration when meaningful—while holding

**Figure 5.** Real application capture of a local prediction-influence view. The current implementation extends this idea with a reconciling waterfall, readable feature values, training-range context, and model-consistent what-if curves.**Figure 6.** Real application capture of the formulation-level explainability workspace and context panel. The panel remains empty until a prediction is available, preventing unrelated global explanations from being presented as local evidence.

all other inputs fixed and calling the same saved model at every point. These curves are therefore genuine model what-if predictions, not interpolated graphics. They are descriptive sensitivity analyses and should not be called causal effects.

11. Formulation efficacy classification

Exact shelf-life regression and efficacy classification answer different questions. The classifier predicts whether a *complete treatment formulation* produces Low, Medium, or High shelf-life improvement relative to a matched control.

Every treatment row is joined to the control row sharing the same `context_id`, and relative improvement is computed as

$$\delta = 100 \frac{y_{treated} - y_{control}}{y_{control}}. \quad (12)$$

Only the training split determines the two tertile thresholds that define Low/Medium/High. Validation and test therefore cannot influence the class boundaries. This is an important leakage-control choice.

The classifier pools V7 categories/tasks because its question is different from the specialist regression problem. It trains a Random Forest classifier and an XGBoost classifier using the same canonical concentration representation. The production

classifier is chosen by validation macro-F1, not test accuracy. Macro-F1 is appropriate because it weights Low, Medium, and High classes equally rather than allowing a dominant class to define the score.

Per-prediction explanations use SHAP for the predicted class. Because Random Forest and XGBoost SHAP outputs are expressed on different raw scales (probability versus margin/logit behavior), the UI avoids displaying misleading cross-model raw magnitudes and instead reports readable supporting/opposing factors and qualitative strength.

12. Ingredient efficacy ranking

Ingredient ranking asks a third question: “which ingredient tends to be effective across contexts?” A simple descriptive mean is transparent but can be confounded if an ingredient was disproportionately tested under favorable matrices or storage conditions. The system therefore reports both a descriptive effect and a regression-adjusted effect.

Each treated row is first matched with its control and converted to relative improvement δ . The primary ranking fits a ridge-regularized fixed-effects model

$$\delta \sim \text{ingredient} + \text{category} + T + \text{concentration} + pH + a_w + \text{packaging} + \text{application}. \quad (13)$$

with canonical concentration unit also represented categorically. Numeric covariates are standardized before ridge fitting, and the regularization coefficient is $\alpha = 8$. The ingredient coefficient is interpreted as the ingredient’s context-adjusted typical effect within this modeling framework.

The descriptive mean, standard deviation, median, range, and sample count remain visible for transparency. Ingredients with fewer than 20 training observations are flagged low-confidence rather than deleted. Validation- and test-split descriptive means are recomputed independently as a stability check, and the system can compare adjusted ranking with descriptive ranking rather than assuming agreement.

13. AI assistant and application serving

The assistant is not a fifth scientific model. It is a constrained tool-calling interface over the same prediction, classification, ranking, metric, and explanation services used by the UI. The configured primary provider is `gemini-flash-lite-latest`; provider fallbacks can be used only for infrastructure/quota failures. The assistant system prompt explicitly forbids inventing project-specific numbers. Questions that require a metric, prediction, ranking, or explanation must call the corresponding service; narrow deterministic questions can be answered directly from the same in-memory project metadata without an LLM call.

The frontend is implemented in Next.js and the backend in FastAPI. In the deployed architecture, the browser uses same-origin frontend API paths that are rewritten server-side to the backend. Saved model artifacts are loaded for serving; training scripts are not invoked by HTTP requests. This separation makes the dashboard a read-only view of validated artifacts rather than an implicit training environment.

14. Scientific interpretation, limitations, and future work

The main conclusion is intentionally asymmetric. The general shelf-life task is strongly supported internally: three independent category specialists achieve held-out R^2 between 0.920 and 0.971 with stable validation/test behavior. These models are appropriate to keep as the primary deployed research regression system.

Safety is different. Soft/safety contains meaningful predictive signal, but semi-hard/safety and especially hard/safety cannot be defended as precise exact-time regressors. Their weak performance coincides with severe right censoring. The next methodological step should therefore be a censoring-aware model—for example survival/time-to-event analysis, accelerated failure-

time models, or other censored regression formulations—rather than attempting to force a better R^2 through target multipliers or post-hoc calibration [7].

A second limitation is domain shift. Internal context-safe test performance measures generalization within the constructed V7 distribution. Independent literature cases remain a separate, stricter validation layer and should never be used to tune the production model after inspection. Future work should enlarge exact challenge-endpoint evidence, expand coverage of under-represented physical forms and food matrices, and evaluate calibration and error by scientifically meaningful subgroups.

A third limitation concerns causal interpretation. Feature importance, SHAP contributions, and what-if curves explain model behavior; they do not prove that changing a feature experimentally will cause the predicted change. The platform is best described as a research decision-support tool, not a substitute for microbial challenge studies, regulatory validation, or controlled shelf-life experiments.

15. Conclusion

Shelf-Life Studio combines a strong tabular modeling strategy with explicit scientific safeguards. The six-specialist design prevents soft, semi-hard, hard, general, and safety contexts from being collapsed into one global function. Context-grouped splitting reduces leakage; validation rather than test selects the production model; support assessment distinguishes interpolation from extrapolation; uncertainty is calibrated from validation residuals; and local explanations are tied to the exact specialist/model that produced the prediction.

The V7 results provide a clear and defensible story. General shelf-life prediction is strong across all three cheese categories, with LightGBM selected in each specialist and held-out R^2 of 0.971, 0.920, and 0.932. Safety endpoints expose a different statistical structure: censoring increases to 69%, 93%, and 100%, while regression performance decreases correspondingly. Rather than hiding that finding, the application preserves the models as exploratory safety components with explicit caveats and identifies censoring-aware survival modeling as the natural next research step. The complementary formulation classifier, ingredient ranking, explainability, and assistant layers then turn these trained artifacts into an interpretable research workflow rather than a single opaque prediction number.

References

- [1] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, pp. 5–32, 2001.
- [2] G. Ke et al., “LightGBM: A Highly Efficient Gradient Boosting Decision Tree,” in *Advances in Neural Information Processing Systems*, 2017.
- [3] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proc. ACM SIGKDD*, 2016.
- [4] Y. Lou, R. Caruana, J. Gehrke, and G. Hooker, “Accurate Intelligible Models with Pairwise Interactions,” in *Proc. ACM SIGKDD*, 2013.
- [5] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems*, 2017.
- [6] G. Shafer and V. Vovk, “A Tutorial on Conformal Prediction,” *Journal of Machine Learning Research*, vol. 9, pp. 371–421, 2008.
- [7] D. R. Cox, “Regression Models and Life-Tables,” *Journal of the Royal Statistical Society: Series B*, vol. 34, no. 2, pp. 187–220, 1972.